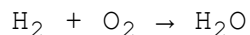


Chemistry — Fraction Balancing Drill

ANSWER KEY · Unit 6.1

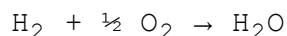
Technique: Save O₂ for last. If you need an odd O count, write a fractional O₂ coefficient, then multiply every coefficient by the denominator to clear it.

WORKED EXAMPLE

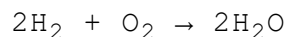


Step 1. Balance H → 1 H₂O gives 2 H ✓

Step 2. Right has 1 O, O₂ supplies 2 → write 1/2 O₂:



Step 3. × 2 to clear the fraction:



Check: H = 4 = 4 ✓ O = 2 = 2 ✓

Solutions

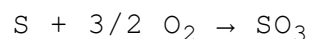
Q1. Practice — same move

5 pts



Step 1. Balance S → 1 SO₃ per S ✓

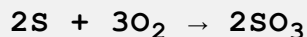
Step 2. Right has 3 O → need 3/2 O₂:



Step 3. × 2 to clear:

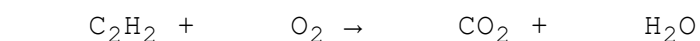
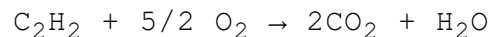
Check: S = 2 = 2 ✓ O = 6 = 6 ✓

FINAL ANSWER



Q2. Build up (acetylene)

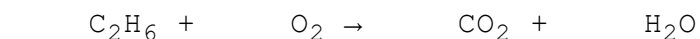
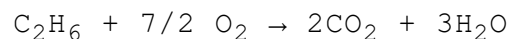
7 pts

**Step 1.** Balance C $\rightarrow 2 \text{CO}_2$ **Step 2.** Balance H $\rightarrow 1 \text{H}_2\text{O}$ **Step 3.** O on right = $2(2) + 1 = 5 \rightarrow$ need $5/2 \text{O}_2$:**Step 4.** $\times 2$ to clear:

Check: C = 4 = 4 $\checkmark$ H = 4 = 4 $\checkmark$ O = 10 = 10 $\checkmark$
FINAL ANSWER

**Q3. Full combustion (ethane)**

8 pts

**Step 1.** Balance C $\rightarrow 2 \text{CO}_2$ **Step 2.** Balance H $\rightarrow 3 \text{H}_2\text{O}$ **Step 3.** O on right = $2(2) + 3 = 7 \rightarrow$ need $7/2 \text{O}_2$:**Step 4.** $\times 2$ to clear:

Check: C = 4 = 4 $\checkmark$ H = 12 = 12 $\checkmark$ O = 14 = 14 $\checkmark$
FINAL ANSWER

